KATAMGARI YAKOBU

Senior Data Engineer | Databricks Lakehouse | GenAI & LLM Data Pipelines

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Open to Relocating

PROFESSIONAL SUMMARY:

- Data Engineer with 5+ years designing and operating production-grade ETL/ELT pipelines and governed data products on the Databricks Lakehouse.
- Deep hands-on expertise in Medallion (Bronze/Silver/Gold) architecture with Delta Lake, Delta Live Tables, and Unity Catalog.
- Advanced PySpark, Spark SQL, and Python engineer with strong orchestration, parameterization, error handling, restartability, and cost optimization.
- Builds RAG (Retrieval-Augmented Generation) ingestion pipelines with chunking, cleaning, and embeddings that power internal LLM knowledge assistants.
- Hands-on with embeddings and vector search across Databricks Vector Search, FAISS, Pinecone, and ChromaDB, with LangChain/LlamaIndex orchestration.
- Integrates LLM enrichment via OpenAI, Azure OpenAI, Databricks Mosaic AI / Foundation Model APIs, and Hugging Face into batch and streaming products.
- Delivers machine learning (XGBoost, scikit-learn, Random Forest, regression) for demand forecasting and anomaly detection, tracked end to end in MLflow.
- Engineers real-time streaming with Apache Kafka and Spark Structured Streaming, cutting data latency from hours to minutes.
- Implements data-quality, validation, and reconciliation frameworks (Great Expectations) with automated tests, alerting, lineage, and Unity Catalog governance.
- Optimizes Spark and SQL via partitioning, caching, Z-ordering, and Delta OPTIMIZE, and modernizes legacy IBM DataStage estates onto Databricks.
- Experienced across Azure, AWS, Snowflake, and BigQuery with GitHub-based CI/CD, Docker, and Terraform, partnering cross-functionally in Agile delivery.

CERTIFICATIONS:

Microsoft Certified: Data Analyst Associate (Power BI)

IBM Data Analyst Professional Certificate

Alteryx Designer Core Certification

Tableau Desktop Specialist

In Progress / Targeted: Databricks Certified Data Engineer Associate, Databricks Certified Generative AI Engineer Associate, AWS Certified Data Engineer – Associate, SnowPro Core

PROFESSIONAL EXPERIENCE

Senior Data Engineer

(Enterprise Lakehouse & GenAI-Ready Trusted Data Products) Feb 2022 – Aug 2024

Infosys Ltd | Client: Walmart | Bengaluru, India

IT Services & Consulting. Built and modernized Walmart's enterprise Databricks Lakehouse, evolving it from a traditional analytics platform into a governed foundation that also powers Retrieval-Augmented Generation and LLM-driven internal applications.

Responsibilities:

- Designed and built scalable Databricks/PySpark pipelines to ingest, transform, validate, and publish trusted data products from Azure SQL, REST APIs, ServiceNow, relational databases, and flat files, establishing a single governed source of truth that cut report-preparation time by 40%.
- Implemented end-to-end Bronze/Silver/Gold medallion layers with Delta Live Tables and Unity Catalog, standardizing how 20+ downstream teams consumed curated data and reducing duplicate data marts by 35%.
- Engineered a RAG ingestion pipeline that chunked, cleaned, and embedded enterprise policy and product documents into Databricks Vector Search, enabling an internal LLM (Azure OpenAI) knowledge assistant and reducing manual document lookup time for support teams.
- Integrated LLM-based enrichment (entity extraction, classification, summarization) into Silver-layer pipelines using Foundation Model APIs, automating categorization of unstructured ticket and catalog text that previously required manual tagging.
- Developed ETL/ELT workflows with full orchestration, parameterization, error handling, and restartability, lifting pipeline reliability to 99% on-time delivery and eliminating manual reruns that cost ~10 hours/week.
- Re-engineered legacy IBM DataStage and application-to-application integration jobs onto Databricks, modernizing fragile batch flows and reducing integration maintenance effort by 50%.
- Implemented data-quality, validation, and reconciliation frameworks with automated tests and alerting (Great Expectations), surfacing issues before they reached executives and improving SLA adherence for business-critical dashboards.
- Tracked governed datasets and ML features in MLflow and Unity Catalog, providing lineage, versioning, and access controls that satisfied enterprise governance and audit requirements.
- Adopted GitHub-based CI/CD (branching, pull requests, code review, promotion across Dev/Test/Prod), accelerating safe deployments and improving auditability across the data team.
- Partnered with product owners, architects, ML engineers, and analytics teams in Agile delivery to translate business needs into KPIs, semantic structures, and AI-ready data products supporting executive decision-making.
- **Environment:** Databricks, Delta Lake, Delta Live Tables, Unity Catalog, PySpark, Spark SQL, Databricks Vector Search, Azure OpenAI, MLflow, Airflow, GitHub, Power BI.

Data Engineer

(High-Volume Operational Data Integration & Demand-Forecasting Platform) Dec 2019 – Feb 2022

Ventas Softech Pvt Ltd | Client: Amazon | Hyderabad, India

IT Services & Consulting. Built a large-scale streaming and batch Lakehouse that ingested operational data across fulfillment systems and powered machine-learning demand forecasting.

Responsibilities:

- Engineered scalable ETL/ELT pipelines ingesting 1M+ daily order, inventory, and shipment records from SAP ERP, Azure SQL, ServiceNow, and Seller Central APIs into a layered Lakehouse for unified, near-real-time reporting.
- Built Kafka + Spark Structured Streaming ingestion for near-real-time inventory events, reducing data latency from hours to minutes for operations dashboards.
- Developed Python/PySpark transformation and feature-engineering workflows producing clean, model-ready Silver/Gold datasets that fed forecasting models across 15+ fulfillment centers.
- Built, trained, and validated demand-forecasting models (Random Forest, XGBoost, regression) reaching ~91% accuracy, tracking experiments and metrics in MLflow and giving operations teams earlier demand signals that reduced stockout-driven costs.
- Delivered curated, reusable data products with dimensional/medallion modeling, reducing redundant pipelines and improving downstream query performance and usability.
- Automated data validation, quality checks, reconciliation, and monitoring on ingestion jobs, reducing data defects and late-arriving data by 30%.
- Optimized warehouse/Spark performance via partitioning and clustering, cutting query runtimes and lowering compute cost on high-volume tables.
- Orchestrated workflows with Apache Airflow and modular, tested transformation models, improving end-to-end restartability and dependency management.
- Used Git/GitHub for version control, code review, and release management of data-engineering assets, raising code quality and reducing production incidents.
- Collaborated cross-functionally with business partners, data scientists, and analysts to translate requirements into well-documented data models and KPIs, accelerating self-service reporting.
- Built reusable Python/PySpark utility modules and parameterized job templates, cutting new pipeline development time by ~20% across fulfillment center onboarding.
- Set up job monitoring, alerting, and SLA dashboards for critical ingestion pipelines, shortening incident detection and response time for on-call teams.

Environment: Apache Kafka, Spark Structured Streaming, Databricks, Delta Lake, Python, PySpark, Airflow, XGBoost, scikit-learn, MLflow, SAP ERP, Azure SQL, AWS S3, GitHub.

ACADEMIC & PERSONAL PROJECTS

Capstone Project — Enterprise GenAI Knowledge Platform (RAG + LLM)

M.S. Information Studies Capstone | Trine University 2025 – 2026

Designed and built an end-to-end Retrieval-Augmented Generation platform that lets users query thousands of unstructured documents through a natural-language LLM interface, backed by a fully governed Lakehouse data layer.

Tech stack: Databricks, Delta Lake, Unity Catalog, PySpark, LangChain, Hugging Face embeddings, Databricks Vector Search / FAISS, Azure OpenAI (GPT-4), MLflow, Airflow, Python, Streamlit.

- Built a document-ingestion pipeline that parsed PDF, HTML, and DOCX sources, performed cleaning and semantic chunking, and generated embeddings with Hugging Face / OpenAI models stored in Databricks Vector Search and FAISS.
- Implemented the RAG orchestration layer with LangChain, combining vector similarity retrieval with GPT-4 generation and prompt templates, plus guardrails to reduce hallucination and ground answers in source citations.
- Designed a Medallion data model in Delta Lake to manage raw documents, cleaned chunks, embeddings, and query/feedback logs, all governed under Unity Catalog with row-level access controls.
- Orchestrated incremental re-embedding and index refresh with Airflow, and tracked embedding models, prompt versions, and evaluation metrics in MLflow.
- Evaluated retrieval and answer quality with relevance and faithfulness metrics, improving answer accuracy through chunk-size tuning, hybrid (keyword + vector) search, and re-ranking.
- Delivered an interactive Streamlit front end for natural-language Q&A with cited sources, demonstrating a production-style GenAI data architecture end to end.

Mini Project — Real-Time Streaming Data Quality & Anomaly Detection

Self-Directed Data Engineering Project 2025

Built a compact, fully open-source streaming pipeline that detects data-quality anomalies in real time, deliberately using a different stack from the capstone to broaden tooling coverage.

Tech stack: Apache Kafka, Spark Structured Streaming, dbt, Snowflake, Great Expectations, scikit-learn (Isolation Forest), Docker, Python, GitHub Actions.

- Generated and streamed synthetic transaction events through Apache Kafka, consumed with Spark Structured Streaming into a raw landing zone.
- Applied Great Expectations validation suites on streaming micro-batches to catch schema drift, nulls, and range violations, routing failures to a quarantine table.
- Trained an Isolation Forest (scikit-learn) model to flag anomalous transactions in near-real time, surfacing suspected outliers for review.
- Modeled curated marts in Snowflake with dbt, adding tests, documentation, and lineage for downstream analytics.
- Containerized the full pipeline with Docker and automated build/test with GitHub Actions, making the project reproducible end to end.

CORE COMPETENCIES

Data Pipeline Architecture • Lakehouse & Medallion Design • GenAI / RAG Data Engineering • LLM & Embedding Workflows • Streaming & Batch ETL/ELT • ML Feature Engineering • Data Quality & Governance • Performance & Cost Optimization • CI/CD & DataOps • Stakeholder Collaboration & Agile Delivery

TECHNICAL SKILLS:

Lakehouse & Big Data: Databricks (Notebooks, Workflows/Jobs, Delta Lake, Delta Live Tables, Unity Catalog, Photon), Apache Spark, PySpark, Spark SQL, Spark Structured Streaming, Apache Kafka

AI / ML / LLM / GenAI: RAG pipelines, embeddings & vector search (Databricks Vector Search, FAISS, Pinecone, ChromaDB), LangChain, LlamaIndex, Databricks Mosaic AI & Foundation Model APIs, OpenAI / Azure OpenAI, Hugging Face Transformers, MLflow, Feature Store, model serving, prompt engineering

Machine Learning: scikit-learn, XGBoost, Random Forest, regression, time-series forecasting, anomaly detection, feature engineering, model evaluation & monitoring

Languages: Python (Pandas, NumPy, PySpark), SQL (advanced), Spark SQL, Bash, basic Scala

ETL / ELT & Integration: ETL/ELT design, dbt, IBM DataStage, CDC, batch & incremental loading, REST/JSON APIs, application-to-application integration, ServiceNow integration

Orchestration & DataOps: Databricks Workflows, Apache Airflow, dbt, scheduling, dependency management, monitoring/alerting, restartability, observability, data contracts

Data Modeling: Medallion, dimensional / star-schema, semantic layers, curated reusable data products, data warehousing, vector & feature schemas for ML

Cloud & Warehouses: Azure (Data Factory, Azure SQL, ADLS Gen2, Azure OpenAI), AWS (S3, Glue, Redshift, Lambda), Snowflake (incl. Cortex), Google BigQuery

Governance & Quality: Unity Catalog, Great Expectations, validation & reconciliation frameworks, data dictionaries, lineage, metadata, access controls, PII handling

DevOps & Source Control: GitHub (branching, PRs, code review, release mgmt), CI/CD, Docker, environment promotion (Dev/Test/Prod), Terraform (basics)

Analytics & BI: Power BI, Tableau, KPI & executive reporting, operational dashboards

Testing & Documentation: pytest, unittest, data-quality test suites, technical documentation, runbooks, data dictionaries

Collaboration & Delivery: Agile/Scrum, Jira, Confluence, cross-functional stakeholder communication, requirements gathering

EDUCATION:

Master of Science in Information Studies
Trine University, Michigan, USA Jan 2025 – Jul 2026

Bachelor of Technology , PBR VITS, India 2019 – 2021